

SHRUTHI KODATI

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Professional Summary

Full-Stack Software Engineer with 6+ production systems built end-to-end using Python, Flask, Node.js, JavaScript, and SQL, from relational schema design to CI/CD cloud deployment. Delivered 40% reduction in manual processing effort and 35% faster API response times on live systems, with proven ownership across RESTful API architecture, secure authentication, and scalable data pipelines.

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL, C#, Shell

Databases: MySQL, Microsoft SQL Server, MongoDB, PostgreSQL, Redis, Schema Design, Query Optimization, Indexing, Transactions

Frontend & Backend: Flask, Node.js, Express.js, Next.js, React.js, Redux, HTML5, CSS3, Tailwind CSS, RESTful API Design, GraphQL, WebSockets, Socket.io, MVC Architecture, Authentication & Authorization, JWT, OAuth 2.0, OOP

Data & ML: Pandas, NumPy, Scikit-learn, PyTorch, ETL Pipelines, FAISS, BM25, Data Modeling, Stream Processing, Geospatial Analytics

Security: Signal Protocol (X3DH + Double Ratchet), Ed25519, E2E Encryption, Zero-Knowledge Architecture, Replay Attack Prevention

Cloud & DevOps: Git, GitHub, Docker, CI/CD Pipelines, Render, Vercel, Microsoft Azure, AWS (S3, EC2), Linux, Postman, Nginx, Agile, SDLC, Code Review, Unit Testing

Visualization: PowerBI, Tableau, Altair, Plotly, Folium, Jupyter Notebook

Professional Experience

Software Engineer, Graduate Assistant | *University of Illinois Chicago, College of Nursing*

Oct 2025 – May 2026

- Spearheaded greenfield development and cloud deployment of a live production web application using Python, Flask, RESTful APIs, HTML5, CSS3, and JavaScript on Render via Git/GitHub CI/CD, automating end-to-end clinical dietary workflows for 3+ active nursing research teams and eliminating **40% of manual processing effort** across 500+ structured dietary records per workflow cycle.
- Re-architected 12+ RESTful API endpoints by restructuring SQL query logic, introducing indexed relational schemas, and optimizing data pipeline throughput, achieving a **35% reduction in average API response time** and hardening reliability under concurrent production load serving 10+ daily active users.
- Engineered a responsive, accessible frontend with DOM-driven dynamic rendering, client-side form validation, and real-time user feedback across 8+ core workflow views, empowering non-technical clinical staff to confidently interact with complex structured data without engineering support.
- Designed and implemented secure authentication and session management flows, multi-layer server-side input validation, and structured CRUD operations covering **10K+ managed database records**, building the complete application stack to production engineering standards with defensible, maintainable architecture.
- Owned the full CI/CD delivery lifecycle including pipeline configuration, zero-downtime version releases, Git branching strategy, and systematic performance profiling, sustaining **100% production uptime** across 6+ months of continuous active development.
- Developed 6+ dynamic REST API endpoints for real-time data querying, structured aggregation, and clinical report generation, enabling nursing researchers to programmatically filter, access, and export structured dietary datasets directly from the live production system.
- Partnered with nursing researchers and faculty across 10+ requirement sessions to elicit specifications, validate features, and drive rapid iteration cycles, delivering version-controlled zero-downtime production releases that 15+ active clinical users depend on daily.
- Conducted systematic debugging, code reviews, and production performance profiling across the full application stack, proactively resolving 40+ critical bottlenecks and documenting all technical decisions to maintain engineering standards throughout the complete project lifecycle.

Projects

Cipher-Signal Protocol Encrypted Messenger | *Node.js, Socket.io, Signal Protocol, Ed25519*

- Implemented the complete Signal Protocol stack (X3DH key agreement + Double Ratchet algorithm) from scratch in Node.js and Socket.io, producing a **zero-knowledge platform** supporting **25MB encrypted file transfers** across 12 file types with forward secrecy and post-compromise recovery enforced at the protocol layer.
- Hardened security with Ed25519 SPK signature verification, **256-bit session tokens**, rate limiting, and replay attack prevention, eliminating path traversal vulnerabilities via UUID-namespaced storage and guaranteeing **exactly-once message delivery** via persistent offline queue with ACK-based deduplication.

Multi-Modal RAG for SEC 10-K Financial Document Analysis | *Python, Flask, FAISS, BM25, LLaMA 3.2*

- Architected a two-stage RAG pipeline ingesting **150 SEC 10-K filings (8GB, 45K text chunks, 12K tables)** with hybrid FAISS+BM25 retrieval and table row-sentence representation, achieving **85% Table Recall@5** and a **12-point Exact Match improvement** over text-only baselines.
- Deployed the production retrieval system (LLaMA 3.2, sentence-transformers, Flask APIs) with intelligent query routing and citation generation, attaining **96% faithfulness** and **4.2s median latency** across FinQA, DocFinQA, and FinDER benchmarks with full ablation studies.

SF Property Tax Analysis & Gentrification Risk Assessment | *Python, Altair, GeoPandas, REST APIs*

- Constructed an end-to-end analytics pipeline processing **3.7M property records (6GB GeoJSON)** across **93 San Francisco neighborhoods**, implementing memory-optimized ETL with categorical encoding and streaming parsing that slashed data footprint by **87% (5.7GB to 733MB)**.
- Surfaced **59.6% median property appreciation (2015 to 2023)** via **15+ interactive Altair visualizations** identifying high-risk displacement zones and quantifying COVID-19 housing market impact for urban planners and policymakers.

Education

Master of Science in Computer Science | *University of Illinois Chicago* | **GPA: 3.6/4.0**

Aug 2024 – May 2026

Bachelor of Technology in Computer Science | *CVR College of Engineering* | **GPA: 8.08/10**

Aug 2020 – May 2024